

K-FIELD

OVERDETERMINED ONE-DIMENSIONAL SERIALIZED SPECTRAL ADDRESS SPACE

A foundational reference on signed-scalar volumetric addressing, hidden rank-three geometry, scalar group composition, metric recovery, shell-density dimension, direct-reciprocal duality, the locked Mass-Bridge, recursive radix structure, geometric error correction, and discovery pathways.

K-Field as an overdetermined one-dimensional serialized spectral address space

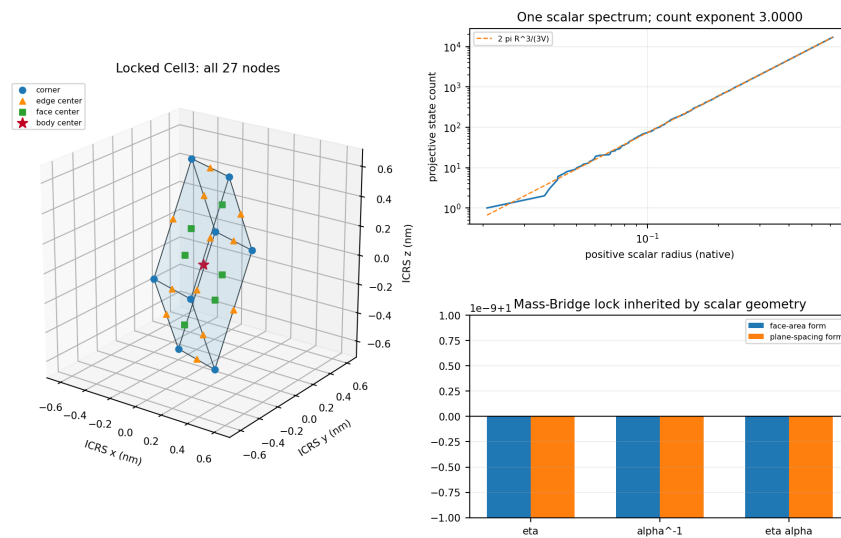


Figure 1. Title composite generated exclusively from the locked Cell3 basis, the computed scalar shell registry, and the exact direct/reciprocal Mass-Bridge identities.

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Frontispiece: dimensionally correct locked Cell3, planes, and complete local grammar

Dimensionally correct locked Cell3: six planes, twelve edges, all 27 addressed nodes

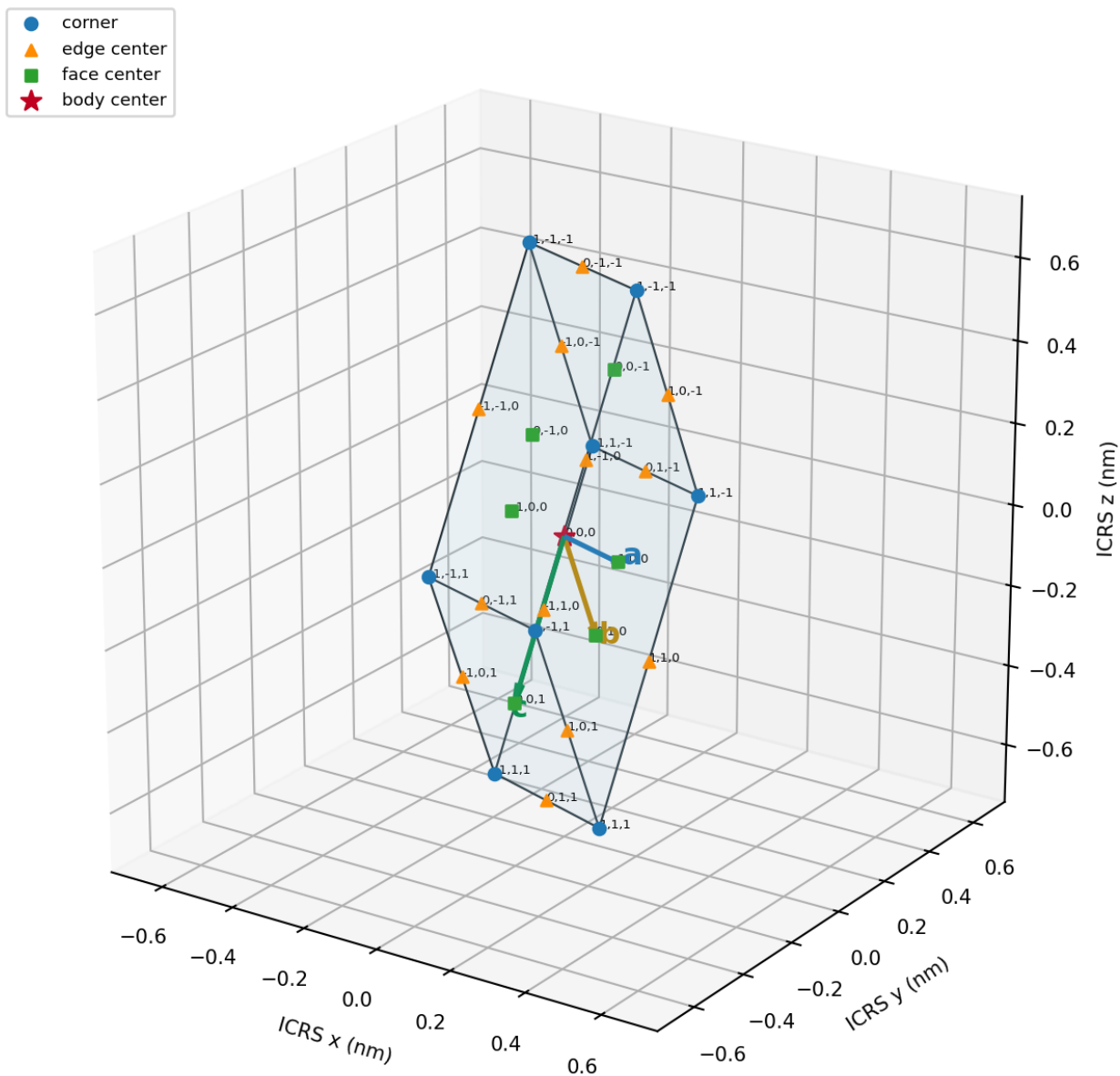


Figure 2. Locked Cell3 in ICRS physical coordinates. The rendering includes all six boundary planes, twelve edges, twenty-seven centered nodes, exact integer addresses, and the three primitive basis directions. Coordinates are in nanometres.

Boundary-plane atlas

Boundary-plane atlas: complete nine-node arrays with native in-plane dimensions

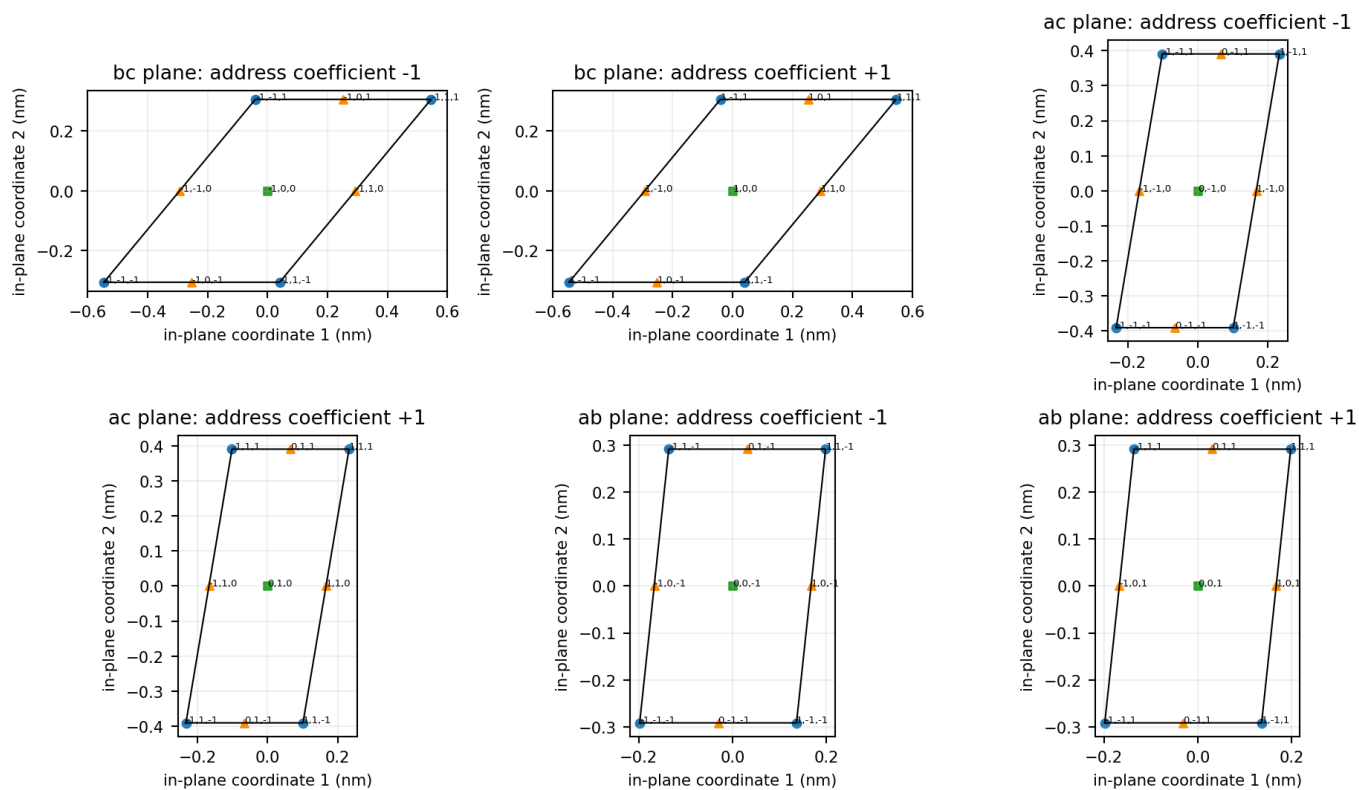


Figure 3. All six boundary planes. Each panel preserves actual in-plane nanometre distances and contains the complete nine-node address array for that plane.

Abstract

The K-Field volume admits a lossless one-dimensional address representation. For every relational address k in Z^3 , the quadratic metric produces $Q(k)=k^T G k$. Under antipodal radial injectivity, equal Q values identify only the pair $\{k, -k\}$. A declared orientation convention therefore converts the nonnegative radius into one signed scalar $\rho(k)$, and ρ uniquely identifies the complete directed volumetric address. The explicit address coordinate is one-dimensional even though its algebraic rank, metric growth, angular differential structure, recursive completion, and Fourier dual retain higher-dimensional organization.

The scalar line is not an ordinary Euclidean line. Its correct addition is induced from integer-vector addition, its correct distance is induced from the Cell3 metric, and its recursive proximity is induced from the 3-adic digit hierarchy. Scalar magnitudes and scalar relational differences reconstruct inner products, angles, face areas, volume, and the full metric. Shell counts grow cubically with radius, so the latent dimension is three even though the explicit address coordinate is one. The direct theta spectrum and its reciprocal Poisson reconstruction supply an analytic fingerprint of dimension, volume, anisotropy, and reciprocal-plane structure.

The locked Mass-Bridge makes this one-dimensional spectrum overdetermined. Its face-area ratios are fixed by $\eta = m_p/m_e$ and α^{-1} . The same lock appears in reciprocal plane spacings and in the diagonal cofactor sector of G^{-1} . The thirteen local projective shell magnitudes simultaneously decompose as $1+6+6$ projective Fourier modes and as six metric coordinates plus seven exact syndromes. At every radix level, a 27-symbol scalar stream carries the complete three-coordinate balanced-ternary address, while a finite carry field preserves rank-three addition. The resulting architecture is a self-indexing spectral state set: one explicit scalar address, a hidden rank-three relational law, a three-dimensional density signature, and heterogeneous geometric redundancy for error detection and correction.

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1. Central theorem and dimensional taxonomy

Let $\Lambda = B^T Z^3 B$ be the Cell3 center lattice, with direct metric $G = B^T B$. A local displacement is indexed by k in Z^3 . Define $Q(k) = k^T G k$. Choose the sign of k by the first nonzero integer component: positive first component gives positive sign and negative first component gives negative sign. The signed scalar address is $\rho(k) = \sigma(k) \sqrt{Q(k)}$.

$$\begin{aligned} Q(k) &= k^T G k \\ \rho(k) &= \sigma(k) \sqrt{Q(k)} \\ Q(k) &= Q(l) \text{ implies } l = \pm k \\ \rho(k) &= \rho(l) \text{ implies } k = l \end{aligned}$$

The first implication is the antipodal radial-injectivity condition. The sign resolves the remaining antipodal ambiguity. Therefore ρ maps the entire discrete volume into a subset S of the real line without loss of directed address identity.

This statement concerns explicit address-coordinate dimension. It does not erase the structures carried by the address. The same object has several dimension counts because each count answers a different question.

Representation	Dimension or split	Meaning
Explicit scalar address	1	One signed real spectral label ρ .
Integer translation module	3	Three independent generators of Z^3 .
Metric state-count growth	3	$N(R)$ grows in proportion to R^3 .
Direction differential	1+2	One radial null mode and two angular modes.
Sheet geometry	1+2	One λ quotient plus a two-dimensional in-sheet fiber.
Symmetric metric response	1+2+3=6	Scale, shape, and orientation of a symmetric 3D metric.
Projective transform	1+6+6	One coherent and two conjugate six-dimensional sectors.
Quadratic channels	6+7	Six metric coordinates and seven compatibility syndromes.
Recursive digit block	1+13x2=27	One mean state and thirteen real phase planes.

The most precise integrated description is: the K-Field is a one-dimensional serialized spectral address set carrying a hidden rank-three relational geometry.

2. Authoritative Cell3 geometry and scale contract

The native basis fixes the dimensionless unit-sphere geometry. The locked physical basis is the same projective shape rendered at nanometre scale, with the exact Mass-Bridge face-area lock. Uniform scale changes lengths, areas, and volume by s , s^2 , and s^3 without changing scalar address identity after normalization or any projective ratio.

Object	a column	b column	c column
Native ICRS x	2.038143851694645e-02	2.198745968323355e-03	-6.303794853473720e-04
Native ICRS y	-3.416414496410549e-03	1.290217025869254e-02	-2.425772201712787e-02
Native ICRS z	-2.332965916728961e-03	-3.406650153605571e-02	-4.292435457109382e-02
Locked nm x	3.275012420558906e-01	3.533077584382840e-02	-1.012931762672882e-02
Locked nm y	-5.489700788400644e-02	2.073198504401918e-01	-3.897877023662599e-01
Locked nm z	-3.748750289472991e-02	-5.474010853885081e-01	-6.897344083672676e-01
Quantity	Native value	Locked physical value	
a	2.079705878243295e-02	3.341796790591105e-01 nm	
b	3.649420513072719e-02	5.864108904111408e-01 nm	
c	4.930856589985942e-02	7.923197650426852e-01 nm	
A_ab	7.547217896726240e-04	1.948691654604841e-01 nm^2	
A_ac	1.011255757688065e-03	2.611062357902053e-01 nm^2	
A_bc	1.385784431733287e-03	3.578095391285615e-01 nm^2	
Volume	2.842066937809649e-05	1.179148919117446e-01 nm^3	
Length anisotropy	2.755690625168	scale invariant	

3. One signed scalar as a complete volumetric address

A conventional scalar address is normally an arbitrary enumeration. The K-Field scalar is generated by the physical metric. Its magnitude simultaneously carries radial shell identity and projective directional identity. Its sign carries the directed antipodal state.

$$\begin{aligned}\Lambda &= B Z^3 \\ p &= o + B k \\ R(k) &= ||B k|| = \sqrt{k^T G k} \\ \rho(k) &= \text{signed } R(k)\end{aligned}$$

Every nonzero address factors uniquely as $k=s u$, where u is primitive and s is a positive integer. Then $R=s \sqrt{u^T G u}$. The primitive ray changes the base scalar and s selects the integer radial level. The scalar therefore encodes both which ray is occupied and which level on that ray is occupied.

$$\begin{aligned}k &= s u, \quad \gcd(|u_1|, |u_2|, |u_3|)=1 \\ R(k) &= s \sqrt{u^T G u}\end{aligned}$$

The 27-state Cell3 as one zero state and 13 signed antipodal scalar pairs

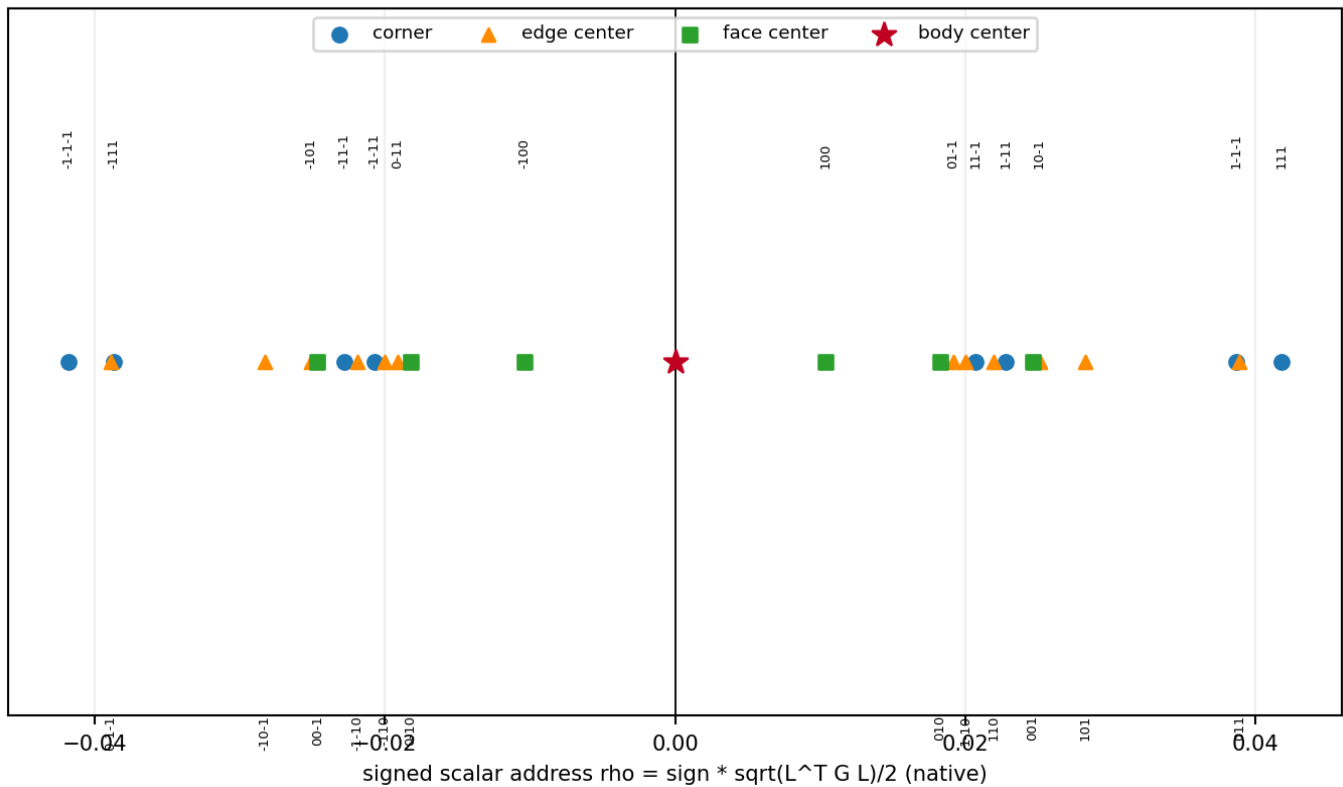


Figure 4. The local 27-state grammar serialized onto one signed scalar axis. The body center is zero. Each of the thirteen positive magnitudes has one negative antipode.

The same dictionary reappears around every lattice center because $\Lambda - Bn_0 = \Lambda$. The scalar address is relational rather than absolute: it identifies the displacement between valid centers, independent of the chosen center used as origin.

4. Hidden group law on the scalar spectrum

The integer address law is additive, but ordinary real addition does not preserve the scalar encoding. In general $\rho(k+l)$ is not $\rho(k)+\rho(l)$. The correct scalar operation is induced from the inverse encoding.

$$\begin{aligned}x (+) y &= \rho(\rho^{-1}(x) + \rho^{-1}(y)) \\ x (-) y &= \rho(\rho^{-1}(x) - \rho^{-1}(y))\end{aligned}$$

With these operations, the scalar spectrum is an Abelian group isomorphic to Z^3 . Translation by a is the nonlinear scalar map $T_a(x) = \rho(\rho^{-1}(x) + a)$. Spatial rank is therefore stored in scalar arithmetic rather than in the number of visible coordinate fields.

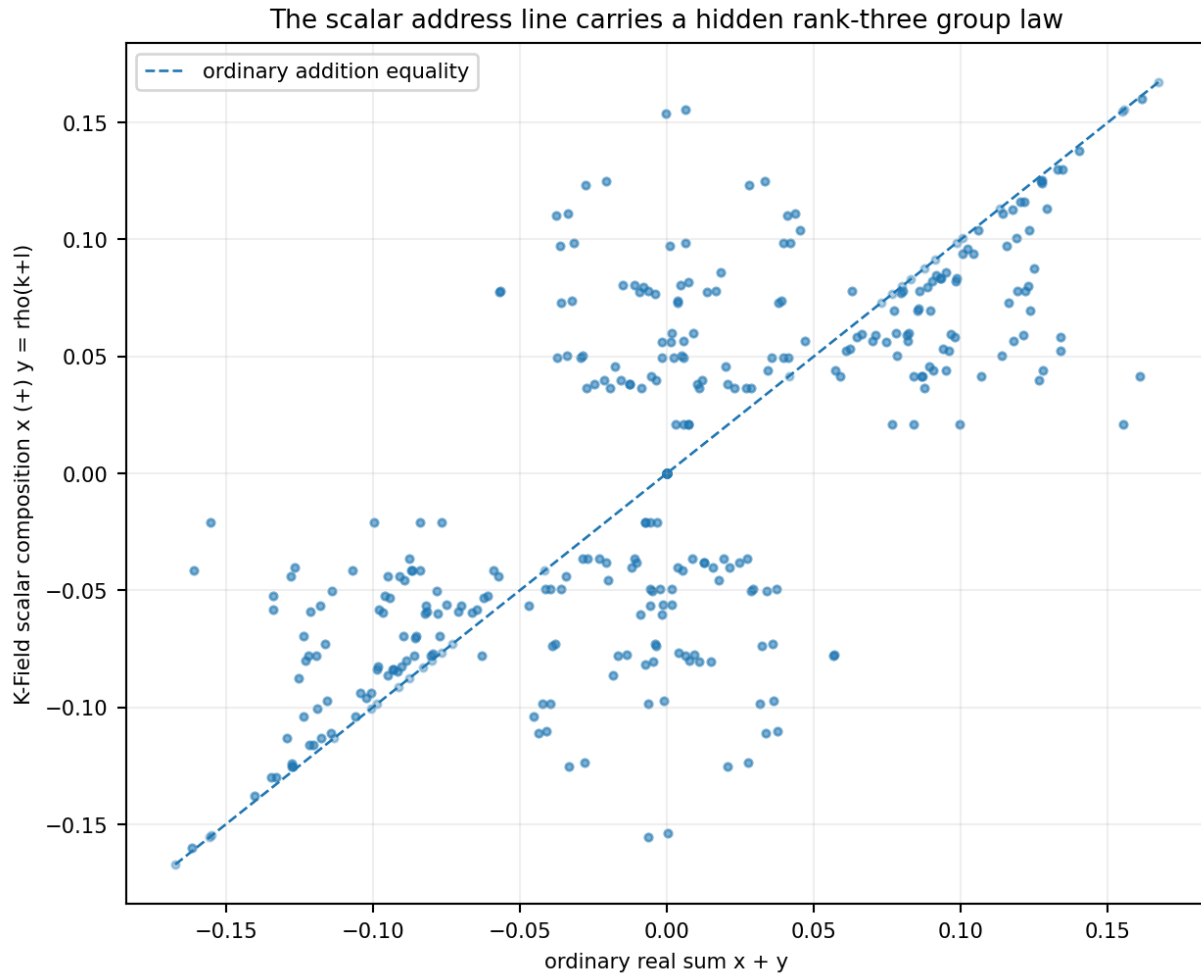


Figure 5. Actual local scalar compositions for all ordered pairs of noncentral one-level Cell3 addresses. Points on the dashed line would obey ordinary real addition. The computed spread shows the nonlinear pulled-back group law.

5. Complete metric recovery from scalar relations

Scalar magnitudes and scalar relational differences reconstruct the metric inner product by polarization. Let $x = \rho(k)$, $y = \rho(l)$, and $x(-)y = \rho(k-l)$. Then

$$\begin{aligned} \langle k, l \rangle_G &= k^T G l \\ &= [Q(k) + Q(l) - Q(k-l)]/2 \\ &= [x^2 + y^2 - (x(-)y)^2]/2 \end{aligned}$$

The angle is therefore recovered from scalar data alone:

$$\cos(\theta_{kl}) = [x^2 + y^2 - (x(-)y)^2] / [2 |x| |y|]$$

For the three primitive axes, the diagonal metric entries are $Q(e_i)$. The off-diagonal entries follow from $Q(e_i - e_j)$. Six scalar shell values determine the complete symmetric metric. Once G is known, one obtains edge lengths, included angles, face areas, volume, reciprocal plane spacings, principal axes, and the entire scalar codebook.

Recovered metric entry	Scalar formula
G11	$Q(e_1)$
G22	$Q(e_2)$

Recovered metric entry	Scalar formula
G33	$Q(e3)$
G12	$[Q(e1)+Q(e2)-Q(e1-e2)]/2$
G13	$[Q(e1)+Q(e3)-Q(e1-e3)]/2$
G23	$[Q(e2)+Q(e3)-Q(e2-e3)]/2$

The thirteen local projective shell values therefore overdetermine the six metric coefficients. Seven independent combinations must vanish. Those combinations are the geometric syndrome sector.

6. Three inequivalent proximity structures

Scalar serialization introduces three distinct notions of closeness. Ordinary scalar proximity is $|x-y|$. Native relational distance is $|x(-)y|$. Recursive proximity is the 3-adic ultrametric determined by the earliest radix level at which the vector addresses differ.

```

d_spectrum(x,y)=|x-y|
d_G(x,y)=|x(-)y|
d_3(k,l)=3^[-min_i v_3(k_i-l_i)]

```

Two states can be adjacent in the sorted scalar spectrum and still be distant in physical space. They can be physically close yet differ at an early recursive digit. They can share a long recursive prefix while occupying substantially different native positions. The scalar line is therefore not a Euclidean replacement for the volume; it is an index set equipped with pulled-back geometric and recursive structures.

Three inequivalent proximity structures around address (10,0,-3)

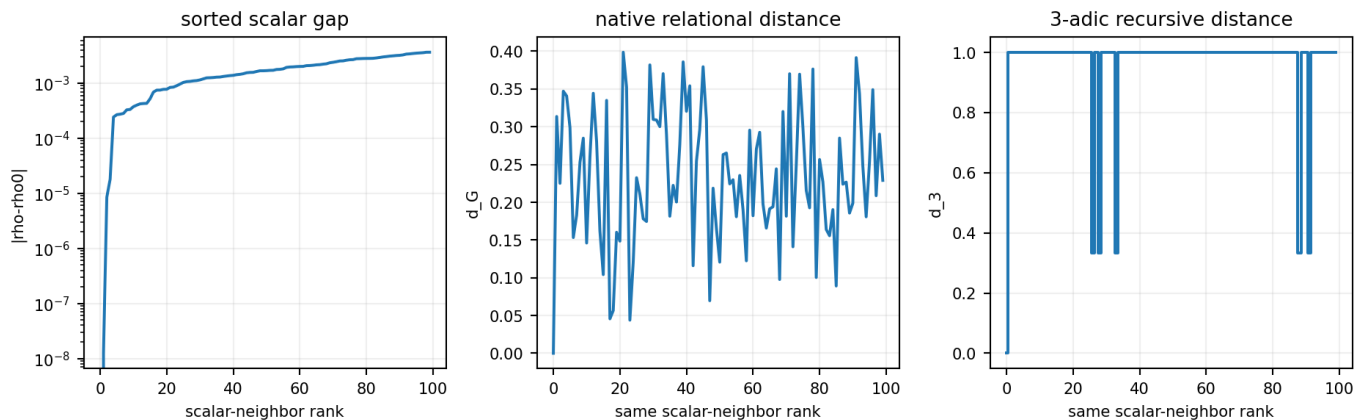


Figure 6. The one hundred closest signed-scalar neighbors of address (10,0,-3), evaluated in three metrics. Scalar rank does not impose the same ordering as native distance or 3-adic distance.

7. Hidden geometric dimension from scalar-state density

The number of positive projective shells below radius R has the asymptotic form $N_+(R)$ approximately $2 \pi R^3/(3V)$. The scalar code is one-dimensional, but its state count grows as the volume of a three-dimensional ball. The latent dimension is therefore recovered by the logarithmic slope $d \log N / d \log R$.

```

N_+(R) ~ 2 pi R^3/(3V)
N_signed(R) ~ 4 pi R^3/(3V)
D_eff = d log N / d log R -> 3

```

Using the complete-sphere region of the computed projective registry with integer bound 30, the fitted count exponent is 2.99996886. The corresponding binned raw scalar-gap exponent is -1.99377162, approaching the expected -2 law.

The volume is recoverable from the scalar spectrum:

$$V = \lim_{R \rightarrow \infty} \frac{2 \pi R^3}{[3 N_+(R)]}$$

A useful uniformized scalar coordinate is $\xi = \text{sign}(\rho) \frac{2 \pi |\rho|^3}{(3V)}$. Its mean large-radius spacing approaches one projective state. It is invariant under uniform geometric scaling because ρ scales as s and V scales as s^3 .

$$\xi(\rho) = \text{sign}(\rho) \frac{2 \pi |\rho|^3}{(3V)}$$

$$\xi(s \rho, s^3 V) = \xi(\rho, V)$$

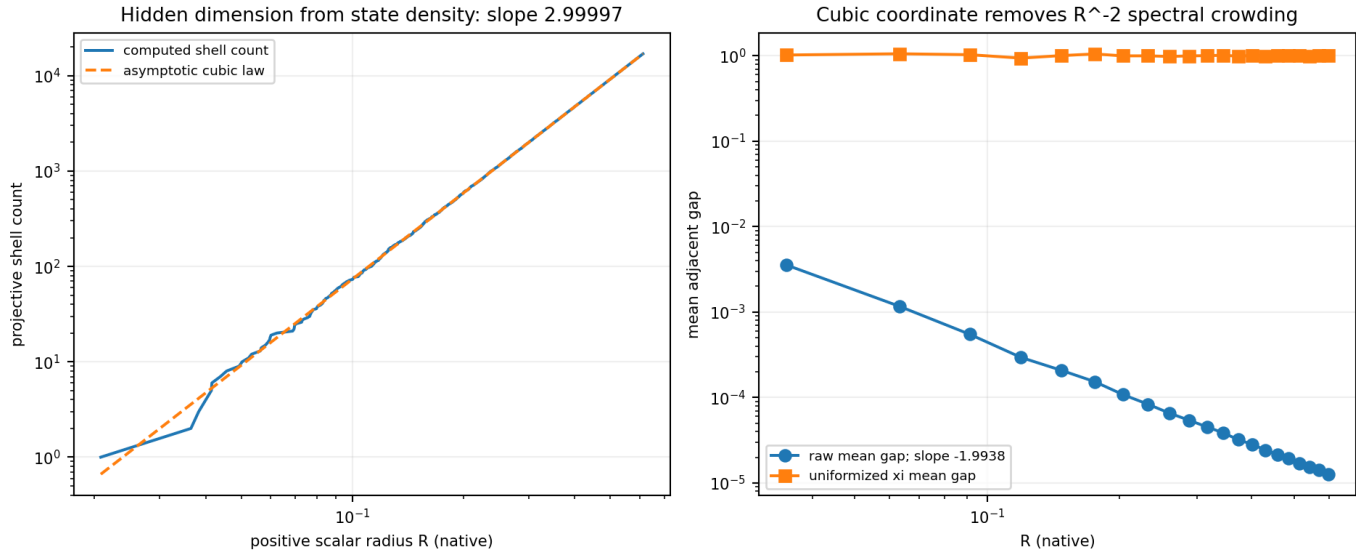


Figure 7. Left: actual cumulative shell count and the cubic asymptotic law. Right: raw spectral gaps contract with radius, while the cubic uniformized coordinate substantially removes the crowding.

8. Theta and zeta spectroscopy of the scalar line

The entire scalar registry is packaged by the theta function $\Theta_G(t) = \sum_k \exp[-t Q(k)]$. Under antipodal radial injectivity, every nonzero magnitude contributes twice.

$$\Theta_G(t) = \sum_{k \in Z^3} \exp(-t k^T G k) \\ = 1 + 2 \sum_{\text{projective } k} \exp(-t \rho(k)^2)$$

Poisson summation reconstructs the same function from the reciprocal metric. The leading factor contains $t^{(-3/2)}$, revealing the latent dimension three, and $1/\sqrt{\det G} = 1/V$, revealing the cell volume. Residual structure carries anisotropy and reciprocal-plane information.

$$\Theta_G(t) = \pi^{3/2} / [t^{3/2} \sqrt{\det G}] \\ * \sum_h \exp[-\pi^2 h^T G^{-1} h / t]$$

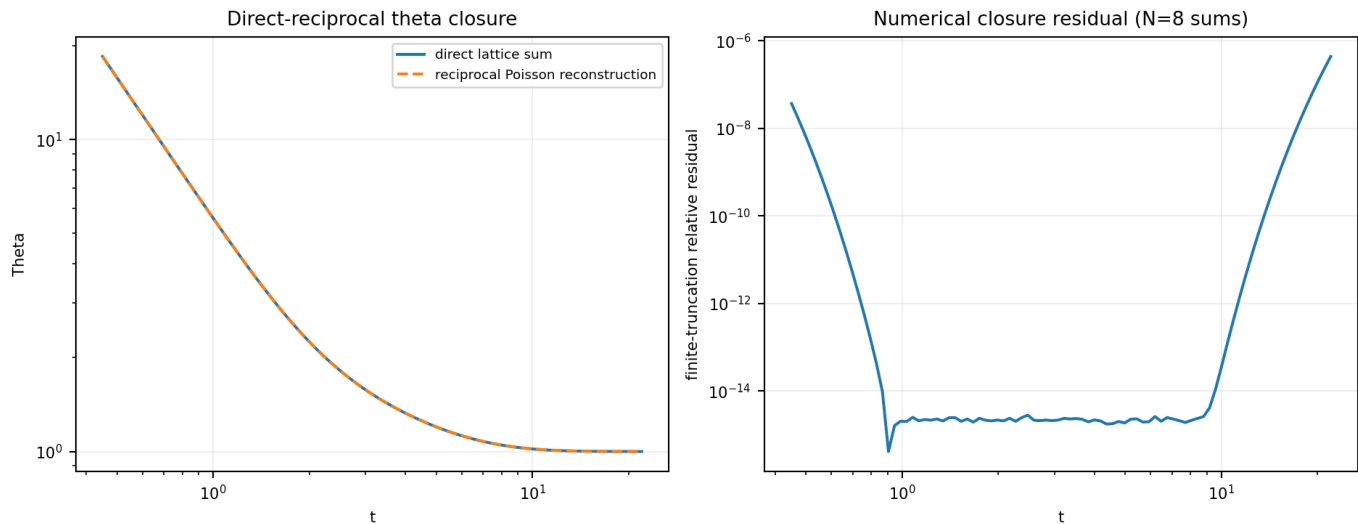


Figure 8. Direct and reciprocal evaluations for the determinant-normalized Cell3 metric. Both curves are computed from finite lattice sums using the supplied metric and its inverse.

The associated Epstein zeta function $Z_G(s) = \sum_{(k \neq 0)} Q(k)^{-s}$ gives a second analytic representation. Its pole structure records dimension, and its functional relation exchanges direct and reciprocal geometry. Theta and zeta spectroscopy provide a rigorous pathway for separating uniform scale, volume, anisotropic shape, and reciprocal response using only the scalar shell sequence.

9. Locked Mass-Bridge as a global scalar-spectrum constraint

The Mass-Bridge is a locked projective ratio system. Its exact face-area conditions are

```
1000 A_bc/A_ab = eta = 1836.152673426000
100 A_bc/A_ac = alpha^-1 = 137.035999177000
10 A_ac/A_ab = eta alpha = 13.399053419929
```

The third relation follows from the first two and removes the shared face A_{bc} . It is the intrinsic projective shape condition. With the established angular and orientation grammar, only uniform scale remains free.

Because $h_a = V/A_{bc}$, $h_b = V/A_{ac}$, and $h_c = V/A_{ab}$, the same lock appears in the reciprocal plane spacings. Because $A_{bc}^2 = V^2 R_{11}$, $A_{ac}^2 = V^2 R_{22}$, and $A_{ab}^2 = V^2 R_{33}$ for $R = G^{-1}$, it also appears in the reciprocal-metric diagonal sector.

```
1000 h_c/h_a = eta
100 h_b/h_a = alpha^-1
10 h_c/h_b = eta alpha

1000 sqrt(R11/R33)=eta
100 sqrt(R11/R22)=alpha^-1
10 sqrt(R22/R33)=eta alpha
```

One locked projective object in direct, reciprocal-plane, and inverse-metric forms

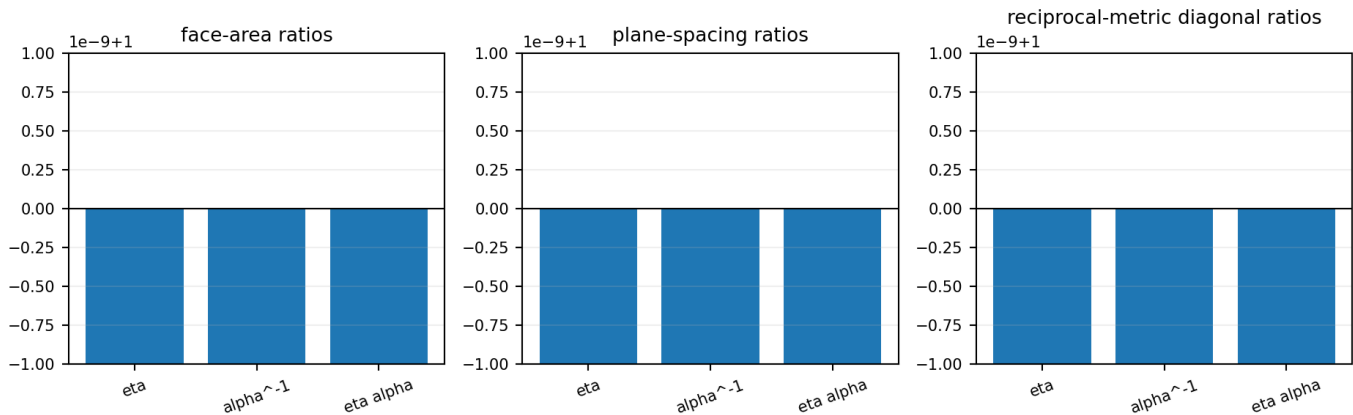


Figure 9. The same locked physical ratios evaluated through face areas, reciprocal plane spacings, and diagonal reciprocal-metric cofactors. Each bar is normalized to its exact target.

Constraint	Direct face form	Plane-spacing form	Reciprocal-metric form	Target
η	1836.152673426000	1836.152673426000	1836.152673426000	1836.152673426000
α^{-1}	137.035999177000	137.035999177000	137.035999177000	137.035999177000
$\eta\alpha$	13.399053419929	13.399053419929	13.399053419929	13.399053419929

Every scalar shell is generated by G. The locked Mass-Bridge therefore constrains the complete one-dimensional spectrum globally. A candidate scalar sequence is not a valid K-Field spectrum merely because its values are distinct; it must reconstruct a metric whose face cofactors and reciprocal planes occupy the locked η - α projective point.

10. The 27-state Cell3 as a signed scalar spectral comb

At one radix level, the address alphabet $\{-1,0,1\}^3$ contains 27 states. The origin is one zero state. The remaining 26 states form thirteen antipodal pairs. Therefore $27=1+13 \times 2$ simultaneously describes the scalar comb, the finite projective channels, and the real character decomposition.

$$\begin{aligned}
 27 &= 1 \text{ zero address} + 13 \text{ positive magnitudes} + 13 \text{ negative magnitudes} \\
 27 &= 1 \text{ mean mode} + 13 \text{ cosine-sine phase planes}
 \end{aligned}$$

The thirteen projective representatives are the points of $PG(2,3)$. Their incidence transform $M=3N-J$ is exactly invertible. Its spectrum contains one eigenvalue -1 , six eigenvalues $+3\sqrt{3}$, and six eigenvalues $-3\sqrt{3}$. The one-dimensional component is coherent across all channels; the two six-dimensional sectors are conjugate projective response spaces.

The same thirteen scalar magnitudes also form a rank-six quadratic metric image with a seven-dimensional null or syndrome space. Hence the same one-dimensional local spectrum supports two exact decompositions: $1+6+6$ for projective response and $6+7$ for metric reconstruction and consistency testing.

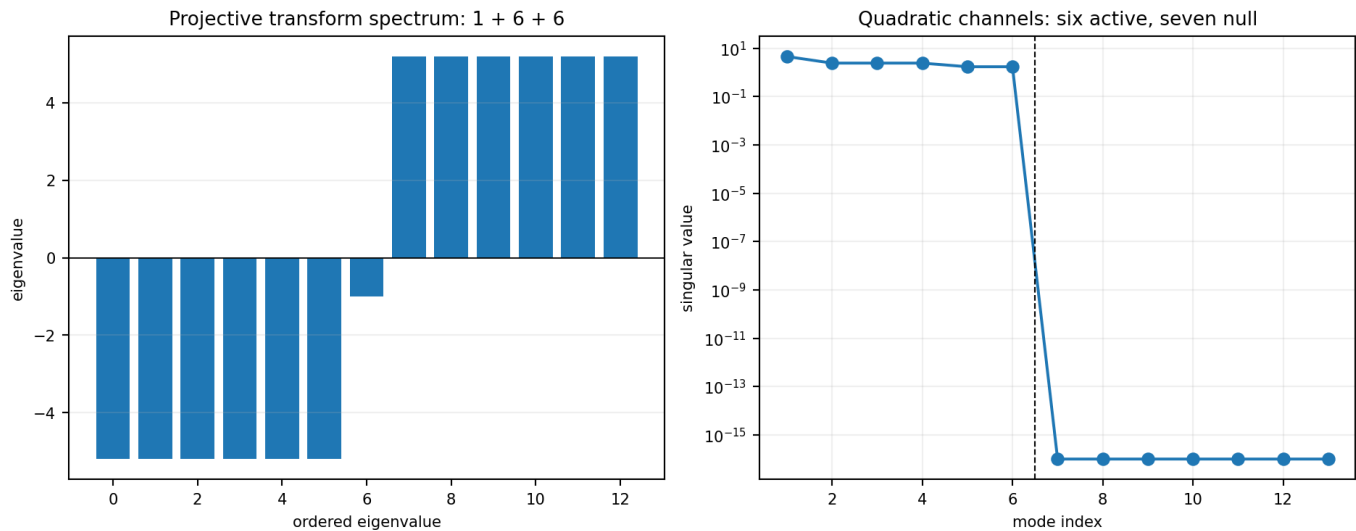


Figure 10. Left: actual eigenvalues of the supplied thirteen-channel projective transform. Right: singular values of the actual thirteen-by-six quadratic design matrix, showing six active dimensions and seven exact null dimensions.

11. Recursive scalarization and the hidden carry field

Every integer address has a unique balanced-ternary expansion $k = \sum_j 3^j L_j$ with vector digits L_j in $\{-1, 0, 1\}^3$. Each vector digit is one of 27 local symbols and can therefore be serialized as one base-27 symbol. At depth d , the finite address group has 27^d states.

$$k = \sum_j 3^j L_j$$

$$L_j \text{ in } \{-1, 0, 1\}^3$$

$$A_d = (Z/3^d \mathbb{Z})^3, \quad |A_d| = 27^d$$

Serialization does not replace the rank-three addition law with ordinary base-27 arithmetic. Digitwise addition requires a three-component carry field:

$$L_j^{(1)} + L_j^{(2)} + C_j = L_j^{(\text{sum})} + 3 C_{(j+1)}$$

All cross-scale nonlinearity introduced by the scalar symbol stream is concentrated in the carry field. The external representation is one-dimensional, while the internal carry dynamics preserve the three coordinate generators. The inverse limit is $\mathbb{Z}_3^3$, which supplies an ultrametric recursive completion rather than an ordinary real-line topology.

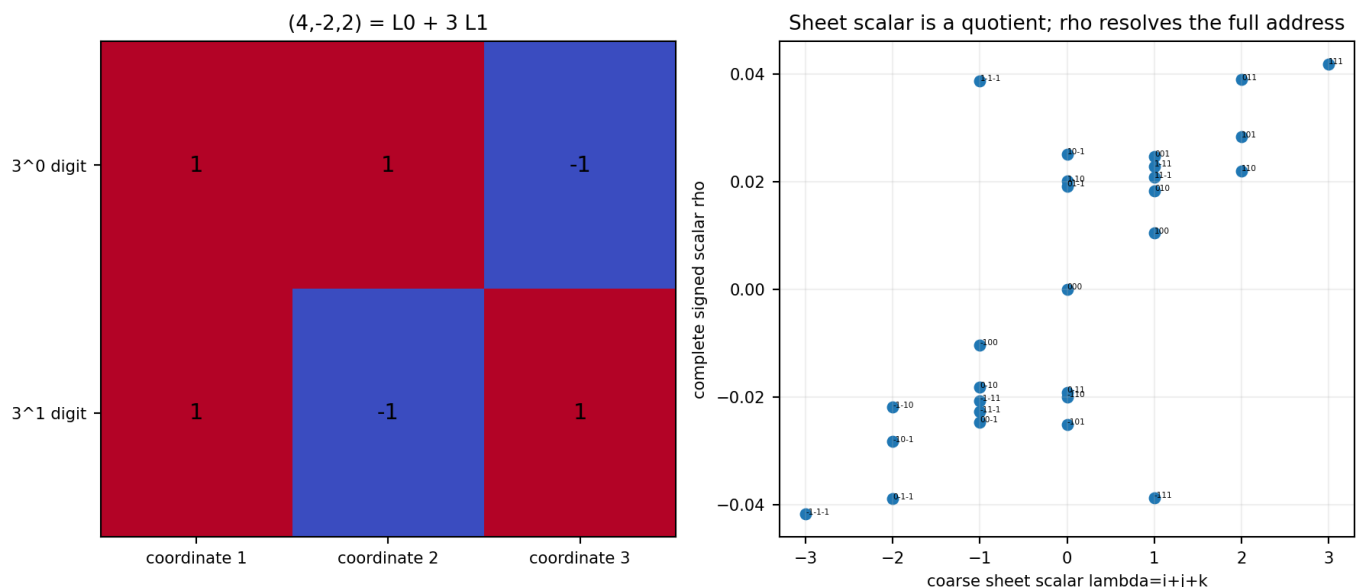


Figure 11. Left: exact balanced-ternary digits for address (4,-2,2). Right: the seven-valued sheet scalar lambda compared with the complete signed scalar rho for all 27 local states.

12. The sheet scalar and complete scalar are different objects

The sheet index $\lambda = i + j + k$ is a signed scalar quotient. It partitions the 27 local states into seven planes with populations 1,3,6,7,6,3,1. It does not uniquely identify a node because multiple addresses share one lambda.

The real address decomposes as $k = \lambda + t + \kappa$, with $1^T \kappa = 0$, where t is the minimum-energy unit-sheet representative. The metric separates exactly: $Q(k) = \lambda^2 d_{\lambda^2 + \kappa^T G \kappa}$. Thus the sheet system is one quotient coordinate plus a two-dimensional in-sheet fiber.

$$\begin{aligned} t &= G^{-1} \mathbf{1} / (\mathbf{1}^T G^{-1} \mathbf{1}) \\ k &= \lambda + t + \kappa, \quad \mathbf{1}^T \kappa = 0 \\ Q(k) &= \lambda^2 d_{\lambda^2 + \kappa^T G \kappa} \end{aligned}$$

The complete scalar rho differs fundamentally: under radial injectivity, it resolves the entire address. A practical hierarchy is therefore lambda for coarse sheet and parity validation, rho for complete address identity, and the reconstructed kappa for in-sheet geometry.

13. Translation-invariant physics becomes scalar-channel physics

For any inversion-even translation kernel $K(k) = K(-k)$, radial injectivity makes K a scalar shell function: $K(k) = F(Q(k)) = F(\rho^2)$. A directed kernel separates into even and odd scalar functions, with the sign of rho selecting the directed branch.

$$\begin{aligned} K_{\text{even}}(k) &= F(\rho(k)^2) \\ K(k) &= F_{\text{even}}(\rho^2) + \text{sign}(\rho) F_{\text{odd}}(\rho^2) \end{aligned}$$

This reduction is lossless because one occupied radius identifies one projective displacement generator. Equal-radius directional mixing is absent in the supplied address registry. Green functions, diffusion laws, elastic kernels, graph Laplacians, interaction Hamiltonians, and response functions can therefore be stored as one scalar coefficient per shell while retaining address specificity.

Convolution on the scalar spectrum must use the hidden operation rather than ordinary subtraction:

$$\begin{aligned} (Af)(x) &= \sum_{y \in S} K(y) f(x(-)y) \\ \chi_{\theta}(x) &= \exp[i \theta \cdot \rho^{-1}(x)] \end{aligned}$$

The explicit address base is one-dimensional, but its harmonic dual remains the three-dimensional character space of Z^3 . This is a direct separation between address-coordinate dimension and phase-space dimension.

14. Noise immunity and the overdetermined scalar codeword

The signed scalar is the minimal address label. Robust decoding uses the full geometric codeword: primitive ray, integer level, antipodal sign, seven direct syndromes, seven reciprocal syndromes, direct-reciprocal inverse closure, sheet and parity state, and global orientation closure.

A noisy witness can move continuously while the decoded address remains exactly fixed inside the closure basin of one valid state. If the observation becomes inconsistent, the decoder returns an invalid witness or an ambiguity rather than forcing the nearest scalar. A genuine transition requires coherent closure on another complete address.

$$\begin{aligned} \Pi_C(p_k + \eta) &= A_k \quad \text{for } \eta \text{ inside closure basin } D_k \\ \partial A_{\hat{}} / \partial \eta &= 0 \quad \text{inside } D_k \end{aligned}$$

Raw scalar spacing controls spectral resolution, but the complete covariance-whitened codeword distance controls address immunity.

$$d_{\text{code}}^2(i, j) = [C(k_i) - C(k_j)]^T \Sigma^{-1} [C(k_i) - C(k_j)]$$

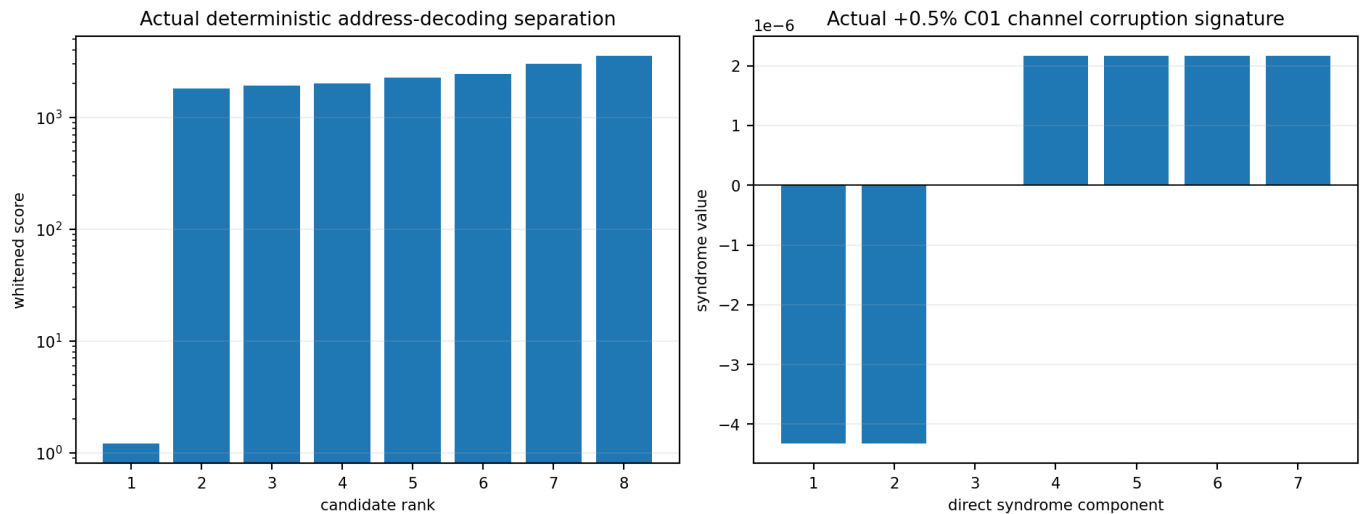


Figure 12. Left: actual deterministic candidate scores for the supplied worked address-decoding calculation. Right: the seven-component syndrome generated by the supplied +0.5 percent C01 channel corruption.

Within the computed bound 30, the closest adjacent positive scalar pair is [4, -8, 21] and [10, 2, 16], with radii 8.956168735962076e-01 and 8.956168736787015e-01, separated by 8.249390059944517e-11 native units. This scalar gap is only one component of the complete closure margin.

15. Path history as a fiber over the scalar address base

A path is an ordered sequence of steps. Its endpoint depends only on the vector sum, so many transport histories can share one integer endpoint and one scalar address. For a monotone primitive-axis route to (i,j,k) , the shortest-path count is the multinomial coefficient. Address (4,-2,2) has $8!/(4!2!2!)=420$ shortest orderings.

$$N_{\text{paths}} = (|i| + |j| + |k|)! / (|i|! |j|! |k|!)$$

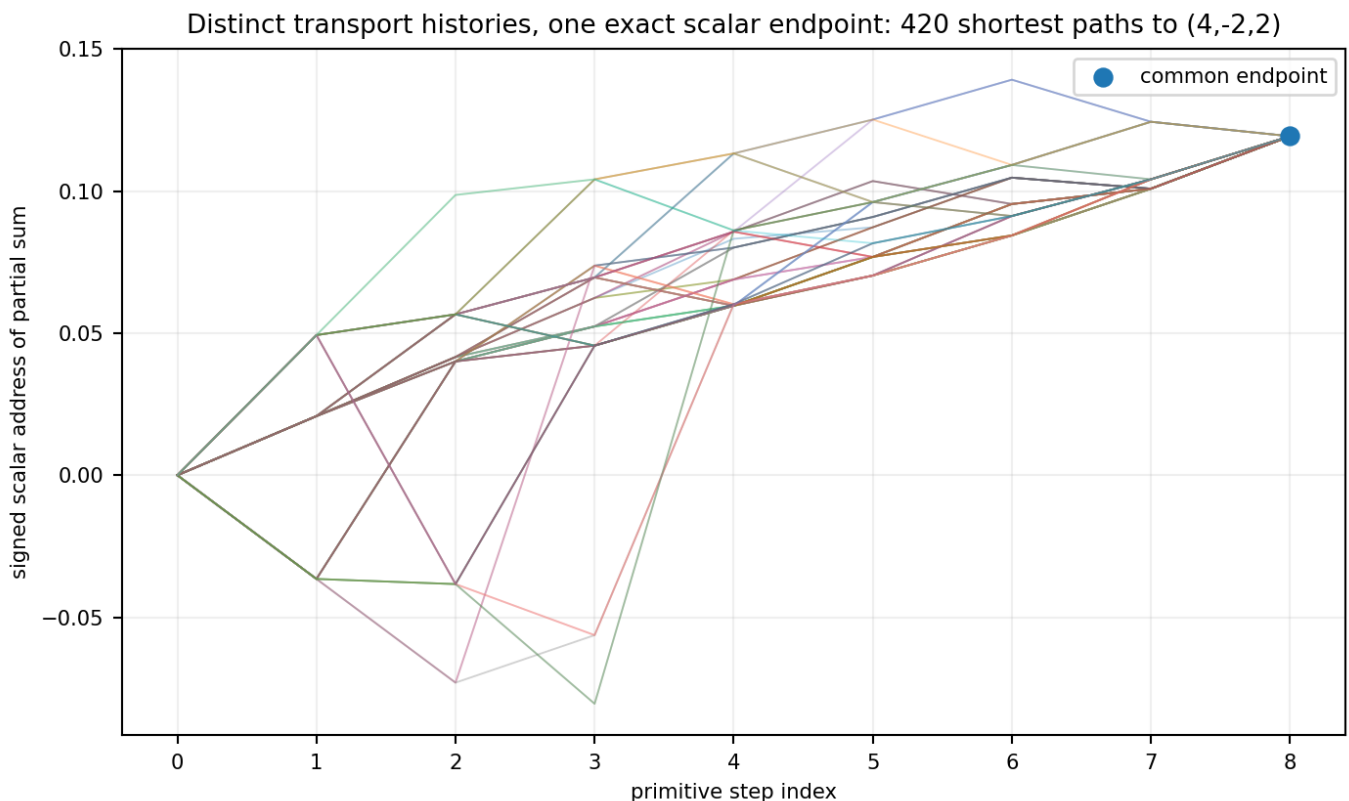


Figure 13. Actual partial-sum scalar trajectories for a representative subset of the 420 shortest paths to (4,-2,2). The histories differ, but all terminate at the same exact scalar address.

Pure Cell3 translation has trivial loop holonomy: a closed integer loop returns to the same scalar address under the pulled-back operation. Any repeatable residual phase, memory, orientation, or impulse around a fully closed address loop is therefore not part of the static scalar address. It is an additional path-dependent state attached as a fiber over the scalar base. This yields a strict experiment: close the scalar address and every geometric syndrome, then test whether an independent phase-like observable also closes.

16. Scalar shell shifts as deformation tomography

A metric perturbation δG shifts the squared scalar shell by $\delta Q_k = k^T (\delta G) k$. The rank-one matrix kk^T is the sensor associated with address k . Six independent shells reconstruct the six components of a symmetric metric perturbation; additional shells measure incompatibility.

$$\delta Q_k = k^T (\delta G) k = \text{trace}[(\delta G) k k^T]$$

$$k \rightarrow k k^T \text{ in } \text{Sym}^2(\mathbb{R}^3)$$

The locked Mass-Bridge adds a global classification. Uniform scale moves every scalar proportionally and preserves all bridge ratios. Orientation changes the rendering but not Q . A general metric deformation changes shell values. A bridge-breaking deformation additionally violates the eta-alpha cofactor constraints. Scalar spectroscopy can therefore separate scale, orientation, allowed deformation, nonmetric corruption, and forced-shape violation.

17. New pathways and consequences

Pathway	Construction	Resulting capability
Scalar spectral atlas	Store ρ , k , λ , parity, projective class, balanced-ternary digits, reciprocal Q , local gaps, and closure margins.	Separates scalar neighbors from native, projective, and recursive neighbors.
Volume-normalized scalar coordinate	Use $\xi = \text{sign}(\rho) 2 \pi \rho ^{3/(3V)}$.	Approximately uniform, scale-invariant projective indexing.
Theta and zeta spectroscopy	Compute $\Theta_G(t)$ and $Z_G(s)$ directly from scalar shells.	Recover dimension, volume, anisotropy, and reciprocal structure.
Global Mass-Bridge syndromes	Add eta and alpha residuals from face, plane, and reciprocal-metric forms.	Detect deformations that violate the forced Cell3 projective point.
Nonlinear scalar arithmetic engine	Implement (+), (-), and translations T_a on the scalar registry.	One-dimensional interface with exact rank-three relational behavior.
Scalar-address compiler	Reversibly map k , ρ , shell rank, radix word, projective channel, and character phase.	Unifies geometry, memory, transforms, and physical readout.
Closure-basin cartography	Compute full whitened codeword margins for a bounded domain.	Produces address reliability and coherent-transition maps.
Holonomy experiment	Close address and all syndromes, then test independent phase closure.	Separates static address geometry from path-dependent dynamics.
Deformation imaging	Invert measured shell shifts through rank-one sensors.	Reconstructs local metric perturbation fields with syndrome diagnostics.

18. Computational reconstruction protocol

Step	Operation
1	Load the locked basis B, direct metric G, reciprocal metric G^{-1} , volume V, and frame transform.
2	Enumerate integer addresses in the declared bounded domain and canonicalize antipodal pairs.
3	Compute $\rho(k)$, primitive factorization, radial level, lambda, parity, reciprocal value, and radix digits.
4	Sort positive rho values to build the scalar codebook and local-gap table.
5	Implement scalar composition through inverse lookup and vector addition.
6	Build the thirteen local projective channels and the exact projective transform $M=3N-J$.
7	Reconstruct G from at least six scalar shells; retain all remaining channels as syndromes.
8	Reconstruct G^{-1} from reciprocal channels and enforce $G G^{-1}=I$.
9	Evaluate the locked Mass-Bridge in face-area, plane-spacing, and reciprocal-cofactor forms.
10	For measured witnesses, generate candidates from scalar and angular intervals, then apply full whitened closure.
11	Return unique address, invalid witness, or unresolved ambiguity; never force an unsupported nearest state.
12	Record raw witness, corrected address, residuals, calibration, and codeword margins for reproducibility.

19. Compact formula ledger

Lattice:	$\Lambda = B Z^3$
Relative address:	$k_{AB} = n_B - n_A$
Composition:	$k_{AC} = k_{AB} + k_{BC}$
Metric:	$G = B^T B$
Quadratic shell:	$Q(k) = k^T G k$
Signed scalar:	$\rho(k) = \sigma(k) \sqrt{Q(k)}$
Scalar composition:	$x (+) y = \rho(\rho^{-1}(x) + \rho^{-1}(y))$
Scalar difference:	$x (-) y = \rho(\rho^{-1}(x) - \rho^{-1}(y))$
Polarization:	$\langle k, l \rangle_G = [x^2 + y^2 - (x(-)y)^2]/2$
Native distance:	$d_G(x, y) = x(-)y $
3-adic distance:	$d_3(k, l) = 3^{-\min_i v_3(k_i - l_i)}$
Shell count:	$N_+(R) \sim 2 \pi R^3 / (3V)$
Mean raw gap:	$\langle \Delta R \rangle \sim V / (2 \pi R^2)$
Uniformized coordinate:	$x_i = \text{sign}(\rho) 2 \pi \rho ^3 / (3V)$
Theta spectrum:	$\Theta_G(t) = \sum_k \exp[-t Q(k)]$
Poisson duality:	$\Theta_G(t) = \pi^{3/2} / (t^{3/2} V) \sum_h \exp[-\pi^2 h^T G^{-1} h / t]$
Balanced ternary:	$k = \sum_j 3^j L_j$
Carry law:	$L1_j + L2_j + C_j = L \sum_j + 3 C_{(j+1)}$
Sheet index:	$\lambda = 1^T k$
Sheet split:	$Q = \lambda^2 d_\lambda^2 + \kappa^T G \kappa$
Metric perturbation:	$\delta Q_k = k^T (\delta G) k$
Mass-Bridge:	$1000 A_{bc}/A_{ab} = \eta$ $100 A_{bc}/A_{ac} = \alpha^{-1}$ $10 A_{ac}/A_{ab} = \eta \alpha$
Plane dual:	$h_a = V/A_{bc}, h_b = V/A_{ac}, h_c = V/A_{ab}$
Reciprocal cofactors:	$A_{bc}^2 = V^2 R_{11}, A_{ac}^2 = V^2 R_{22}, A_{ab}^2 = V^2 R_{33}$
Decoder invariance:	$Pi_C(p_k + \eta) = A_k \text{ inside } D_k$

Appendix A. Direct and reciprocal metric matrices

Matrix row	Column 1	Column 2	Column 3
G native row 1	4.3251765399997141e-04	8.0210431246624869e-05	1.7016744863821204e-04
G native row 2	8.0210431246624869e-05	1.3318270081235947e-03	1.1479192870254278e-03
G native row 3	1.7016744863821204e-04	1.1479192870254278e-03	2.4313346711007794e-03
G ⁻¹ native row 1	2.3775121838982250e+03	3.9628598567349360e-01	-1.6658755027870387e+02
G ⁻¹ native row 2	3.9628598567346474e-01	1.2660574401861390e+03	-5.9777833398345035e+02
G ⁻¹ native row 3	-1.6658755027870387e+02	-5.9777833398345035e+02	7.0518842088287681e+02
G locked nm ² row 1	1.1167605789605009e-01	2.0710333280605807e-02	4.3937235096219773e-02
G locked nm ² row 2	2.0710333280605807e-02	3.4387773239278702e-01	2.9639275896152290e-01
G locked nm ² row 3	4.3937235096219773e-02	2.9639275896152290e-01	6.2777061007729584e-01
G ⁻¹ locked nm ⁻² row 1	9.2080255294575490e+00	1.5348024281925453e-03	-6.4518803569993899e-01
G ⁻¹ locked nm ⁻² row 2	1.5348024281925869e-03	4.9033983123679867e+00	-2.3151755844163890e+00
G ⁻¹ locked nm ⁻² row 3	-6.4518803569993888e-01	-2.3151755844163890e+00	2.7311712747116945e+00

Appendix B. Complete boundary-plane ledger

Family	Coefficient	Unit normal x	Unit normal y	Unit normal z	Spacing nm	Nine node addresses
bc	-1	-0.995965190309	0.083602162507	-0.032619290564	0.329546529695	(-1, -1, -1); (-1, -1, 0); (-1, -1, 1); (-1, 0, -1); (-1, 0, 0); (-1, 0, 1); (-1, 1, -1); (-1, 1, 0); (-1, 1, 1)
bc	1	-0.995965190309	0.083602162507	-0.032619290564	0.329546529695	(1, -1, -1); (1, -1, 0); (1, -1, 1); (1, 0, -1); (1, 0, 0); (1, 0, 1); (1, 1, -1); (1, 1, 0); (1, 1, 1)
ac	-1	-0.089052594102	-0.866576769291	0.491033948326	0.451597379721	(-1, -1, -1); (-1, -1, 0); (-1, -1, 1); (0, -1, -1); (0, -1, 0); (0, -1, 1); (1, -1, -1); (1, -1, 0); (1, -1, 1)
ac	1	-0.089052594102	-0.866576769291	0.491033948326	0.451597379721	(-1, 1, -1); (-1, 1, 0); (-1, 1, 1); (0, 1, -1); (0, 1, 0); (0, 1, 1); (1, 1, -1); (1, 1, 0); (1, 1, 1)
ab	-1	0.194092200808	0.913177168821	0.358379234791	0.605097741519	(-1, -1, -1); (-1, 0, -1); (-1, 1, -1); (0, -1, -1); (0, 0, -1); (0, 1, -1); (1, -1, -1); (1, 0, -1); (1, 1, -1)
ab	1	0.194092200808	0.913177168821	0.358379234791	0.605097741519	(-1, -1, 1); (-1, 0, 1); (-1, 1, 1); (0, -1, 1); (0, 0, 1); (0, 1, 1); (1, -1, 1); (1, 0, 1); (1, 1, 1)

Appendix C. Complete 27-node locked Cell3 ledger

Address	lambda	polarity	class	x nm	y nm	z nm	signed rho native
(-1, -1, -1)	-3	-1	corner	-1.76351350136495 1e-01	1.186824299050373 e-01	6.373114983252528 e-01	-4.18099081171104 5e-02
(-1, -1, 0)	-2	1	edge center	-1.81416008949859 5e-01	-7.62114212780926 6e-02	2.924442941416190 e-01	-2.19360748803017 2e-02
(-1, -1, 1)	-1	-1	corner	-1.86480667763223 9e-01	-2.71105272461222 6e-01	-5.24229100420148 3e-02	-2.07359996406630 7e-02
(-1, 0, -1)	-2	1	edge center	-1.58685962214580 9e-01	2.223423551251331 e-01	3.636109556309988 e-01	-2.83027702812691 8e-02
(-1, 0, 0)	-1	-1	face center	-1.63750621027945 3e-01	2.744850394200322 e-02	1.874375144736496 e-02	-1.03985293912164 7e-02
(-1, 0, 1)	0	1	edge center	-1.68815279841309 7e-01	-1.67445347241126 7e-01	-3.26123452736268 9e-01	-2.51173118974957 5e-02
(-1, 1, -1)	-1	-1	corner	-1.41020574292666 7e-01	3.260022803452290 e-01	8.991041293674473 e-02	-2.28021643378247 3e-02
(-1, 1, 0)	0	1	edge center	-1.46085233106031 1e-01	1.311084291620991 e-01	-2.54956791246889 1e-01	-2.00245087307424 1e-02
(-1, 1, 1)	1	-1	corner	-1.51149891919395 5e-01	-6.37854220210308 2e-01	-5.99823995430522 9e-01	-3.87000069363867 4e-02
(0, -1, -1)	-2	1	edge center	-1.26007291085497 9e-02	9.123392596303405 e-02	6.185677468778878 e-01	3.89197901242903 3e-02
(0, -1, 0)	-1	-1	face center	-1.76653879219142 0e-02	-1.03659925220095 9e-01	2.737005426942540 e-01	-1.82471025653635 9e-02
(0, -1, 1)	0	1	edge center	-2.27300467352786 1e-02	-2.98553776403225 8e-01	-7.11666614893797 8e-02	-1.91528268486241 9e-02
(0, 0, -1)	-1	-1	face center	5.064658813364410 e-03	1.948938511831299 e-01	3.448672041836338 e-01	-2.46542829499297 1e-02
(0, 0, 0)	0	1	body center	0.000000000000000 e+00	0.000000000000000 e+00	0.000000000000000 e+00	0.000000000000000 e+00
(0, 0, 1)	1	-1	face center	-5.06465881336441 0e-03	-1.94893851183129 9e-01	-3.44867204183633 8e-01	2.46542829499297 1e-02
(0, 1, -1)	0	1	edge center	2.273004673527861 e-02	2.985537764032258 e-02	7.116666148937978 e-02	1.915282684862419 e-02
(0, 1, 0)	1	-1	face center	1.766538792191420 e-02	1.036599252200959 e-01	-2.73700542694254 0e-01	1.824710256536359 e-02
(0, 1, 1)	2	1	edge center	1.260072910854979 e-02	-9.12339259630340 5e-02	-6.18567746877887 8e-01	3.891979012429033 e-02
(1, -1, -1)	-1	-1	corner	1.511498919193955 e-01	6.378542202103082 e-02	5.998239954305229 e-01	3.870000693638674 e-02
(1, -1, 0)	0	1	edge center	1.460852331060311 e-01	-1.31108429162099 1e-01	2.549567912468891 e-01	2.002450873074241 e-02
(1, -1, 1)	1	-1	corner	1.410205742926667 e-01	-3.26002280345229 0e-01	-8.99104129367447 3e-02	2.280216433782473 e-02
(1, 0, -1)	0	1	edge center	1.688152798413097 e-01	1.674453472411267 e-01	3.261234527362689 e-01	2.511731189749575 e-02
(1, 0, 0)	1	-1	face center	1.637506210279453 e-01	-2.74485039420032 2e-02	-1.87437514473649 6e-02	1.039852939121647 e-02
(1, 0, 1)	2	1	edge center	1.586859622145809 e-01	-2.22342355125133 1e-01	-3.63610955630998 8e-01	2.830277028126918 e-02
(1, 1, -1)	1	-1	corner	1.864806677632239 e-01	2.711052724612226 e-01	5.242291004201483 e-02	2.073599964066307 e-02
(1, 1, 0)	2	1	edge center	1.814160089498595 e-01	7.621142127809266 e-02	-2.92444294141619 0e-01	2.193607488030172 e-02
(1, 1, 1)	3	-1	corner	1.763513501364951 e-01	-1.18682429905037 3e-01	-6.37311498325252 8e-01	4.180990811711045 e-02

Appendix D. Thirteen projective scalar channels

Channel	Representative	Direct radius	Direct Q/4	Reciprocal Q	Reciprocal magnitude
1	(0, 0, 1)	2.465428294992971e-02	6.078336677751948e-04	7.051884208828768e+02	2.655538402815664e+01
2	(0, 1, 0)	1.824710256536359e-02	3.329567520308987e-04	1.266057440186139e+03	3.558170091755225e+01
3	(0, 1, 1)	3.891979012429033e-02	1.514750063318807e-03	7.756891931021152e+02	2.785119733695690e+01
4	(0, 1, -1)	1.915282684862419e-02	3.668307762933796e-04	3.166802529035916e+03	5.627435054299531e+01
5	(1, 0, 0)	1.039852939121647e-02	1.081294134999929e-04	2.377512183898225e+03	4.875973937479799e+01
6	(1, 0, 1)	2.830277028126918e-02	8.010468055942938e-04	2.749525504223694e+03	5.243591807362291e+01
7	(1, 0, -1)	2.511731189749575e-02	6.308793569560817e-04	3.415875705338510e+03	5.844549345619823e+01
8	(1, 1, 0)	2.193607488030172e-02	4.811913811542040e-04	3.644362196055711e+03	6.036855303927460e+01
9	(1, 1, 1)	4.180990811711045e-02	1.748068416761219e-03	2.820818848414279e+03	5.311138153366262e+01
10	(1, 1, -1)	2.073599964066307e-02	4.299816810975789e-04	5.878282385462897e+03	7.666995751572382e+01
11	(1, -1, 0)	2.002450873074241e-02	4.009809499075791e-04	3.642777052113017e+03	6.035542272333959e+01
12	(1, -1, 1)	2.280216433782473e-02	5.199386984891661e-04	5.210347040405387e+03	7.218273367229442e+01
13	(1, -1, -1)	3.870000693638674e-02	1.497690536876382e-03	3.485583905586401e+03	5.903883387725745e+01

Appendix E. Finite scalar registry certificate

Field	Computed value
Integer bound	$[-30, +30]^3$
Canonical positive projective addresses	113,490
Complete-sphere safe radius	6.126635258663954e-01 native
States within safe radius	16,931
Fitted state-count exponent	2.999968863151
Fitted raw mean-gap exponent	-1.993771618027
Closest address 1	[4, -8, 21]
Closest address 2	[10, 2, 16]
Closest radius 1	8.9561687359620756e-01
Closest radius 2	8.9561687367870146e-01
Closest radial gap	8.2493900599445169e-11
Source finite certificate centers	6,762,624
Source distinct projective shells	3,381,312
Source non-antipodal collisions	0
Source closest delta Q	2.4549251520511461e-12

Appendix F. Source and reproducibility ledger

Source file	Role	SHA-256
K-Field_lossless_discrete_relational_addressing_1784512059(1).json	Lossless relational addressing, scalar shells, projective transform, recursion, theta and metrology	a2ba07f489e4a02ed26ec0bb7f5de5a904c9fd44c3718268179a2497d986b449
K-Field_Geometric_Address_Error_Correction_1784513336(1).json	Direct-reciprocal geometric decoding, channel matrices, worked syndrome calculation	b0ad67c81435fd76480b1307b4f5f50e2c6a6e11463667e36427a93bf657d907
K-Field_functionally_immune_to_non-coherent_noise_in_address_determination_1784513477(2).json	Signed-scalar address theorem, closure basins, functional noise immunity	5d877d4899b06cf95a57528380376b01f7159077662c817670778f549ef63569
K-Field_Mass-Bridge_is_now_a_locked_projective_ratio_system_1785176592.json	Exact Mass-Bridge lock and dimensionally correct physical Cell3	a5ba3ddf21a20e4528c19609a3061fceca60f53f76224f101c9779c24809d7ea

Every numerical figure in this paper is generated from the listed source arrays or from deterministic computations explicitly defined in the formula ledger. No guessed dimensions, illustrative coordinates, or externally sourced graphics are used.